Cognitive Science - Final Project

[redacted]

Section 1. An Intro on Hypothesis Formulating

The hypothesis is a predictive, testable statement concerning the outcome or results that the researcher expects to find from a cognitive/ scientific task. It is a summary of the research question and the independent and dependent variables that are being statistically investigated to stand for the presented theory. One step in formulating a hypothesis, is to study previous research in the area of interest.

Section 2. A Very Brief Literature Review on the Referenced Articles

The general trend of the given articles was as below:

- An emphasis on Repetition Suppression of brain responses and short-term neuronal plasticity in response to stimuli repetition.
- Correlation of repetition suppression and fulfillment of expectation or expectation suppression, and the appearance of significant RS[1] response in the Fusiform Area_among others_ as well as a reduction in this response when stimulated by expected trials rather than surprising.
- Similar effects of RS response in areas such as PPA, more presently for repetitive match trials.

Section 3. The Problem of Interest

Through the fourth project of this course, we investigated the effects of face and nonface recognition responses in PPA and FFA of the cortex. We learned that the FFA was most responsive to human face stimuli while the PPA was more so towards nonhuman stimuli. For this project, and with inspiration from the above articles, we propose the **research question** of investigating the presence of Repetition Suppression in face and nonface stimuli responses invoked in both PPA and FFA under the conditions of matching and non matching trials.

Section 4. Hypothesizing

Considering the previous section's description of the problem, we can now hypothesize a scenario for conducting statistical inference tests and computationally investigating proof for the issue through the datasets. The **Null Hypothesis** can be defined as lack of evidence for the presence of Repetition Suppression, or in other words, the match and

[1] Repetition Suppression

non-match category of trial causing no statistically significant difference in response values in either cases of face _FFA analysis_ or nonface _PPA analysis_ stimuli.

Consequently, the **Alternative Hypothesis** would be defined as evident and statistically significant difference in response values invoked in each of FFA or PPA during stimulation by either matching or non-matching face and nonface stimuli, respectively.

Moreover, it is also notable that the matching and nonmatching set of trials have been considered independent of each other and randomly selected somewhat matching a normal or t distribution. Lastly, the general outlook of the analysis ahead is to consider the two conditions of either matching or non-matching images for each of the face and nonface categories of stimuli and calculate the design matrix of each set of responses. In other words, before running any hypothesis tests, we will be extracting the magnitude of the responses invoked in the related areas of the brain for each category or the Beta weights of the HRF/BOLD signal, to use as inputs of the statistical testing.

Section 5. fMRI Analysis of the Datas

First part of the execution of this project, is to analyze the fMRI data and obtain the Beta weights of the regressors or conditions of interest to be able to perform statistical tests. Since I will be using the SPM tool which I have also employed for the 4th project of this course, here I will only be naming the steps taken and briefly explaining them and have refrained from thorough description of the fMRI analysis.

5. 1. Data Preprocessing

Multiple preprocessing steps were completed through the SPM GUI. These steps were: Realignment, Slice timing, Coregistration, Segmentation, Normalization(or projecting to standardized space), and Smoothing, in order of execution. The output of this batch will be used for the first level analysis, described in 5.2.

5. 2. 1st Level Analysis or the Design of the Generalized Linear Regression Model[2]

In this step, we use the 1st-Level Analysis tool from SPM. In order to use this module, we first need to extract the onsets of each of the conditions from the four given blocks or sessions. As a reminder, we decided that there were two conditions of either matching or non-matching images for each of the face or nonface stimuli, meaning that because we will be analyzing both the PPA(nonface data) and the FFA's(face data) reaction to the matching of stimulus pairs, we will be considering four total conditions of *matching faces, non-matching faces, matching non-faces, and non-matching non-faces* for each run. By entering all the above conditions into the 1st-Level

[2] GLM: Generalized Linear Model

Analysis tool, we will obtain the general design matrix of all the consecutive sessions. This matrix can then be used for the extraction of individual Beta weight vectors for each condition. For further analysis we will define contrasts of non-matching and matching in all the instances of face/nonface stimuli regardless of session, which will finally leave us with 4 independent vectors of Beta that can be used for the next step, or the statistical analysis.

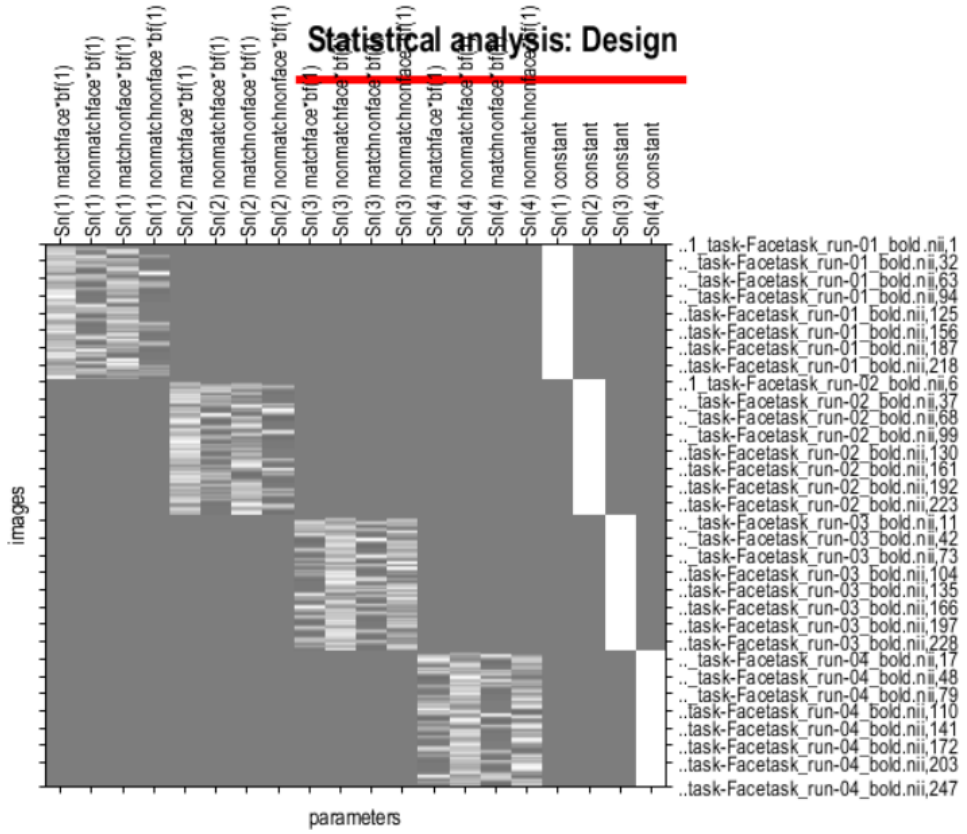


Fig. 1. The obtained design matrix of the model

Section 6. Analysis of the Region of Interest or Masking Methods

After defining the contrasts that we want to investigate, we go ahead and design two separate masks through the WFU-pickAtlas toolbox for individual analyses of the PPA and FFA. For FFA analysis, or investigating the RS responses for the matching and nonmatching face images, we have designed a mask including the BA37 and BA38 atlases. For the PPA analysis, or investigating the RS responses for the matching and nonmatching house/nonface images, we have designed a mask including the BA27, BA28, BA35, and BA36 atlases. [3]

[3] https://www.simplypsychology.org/brodmann-areas.html

We can then use the designed masks as an overlay for the calculated design matrix of the previous section, or the estimated GLM model (SPM.mat), to extract the activity of the desired areas per condition. In figures 2, 3, 4, and 5, we can see the results of this section as displayed over the normalized anatomical images of the brain.

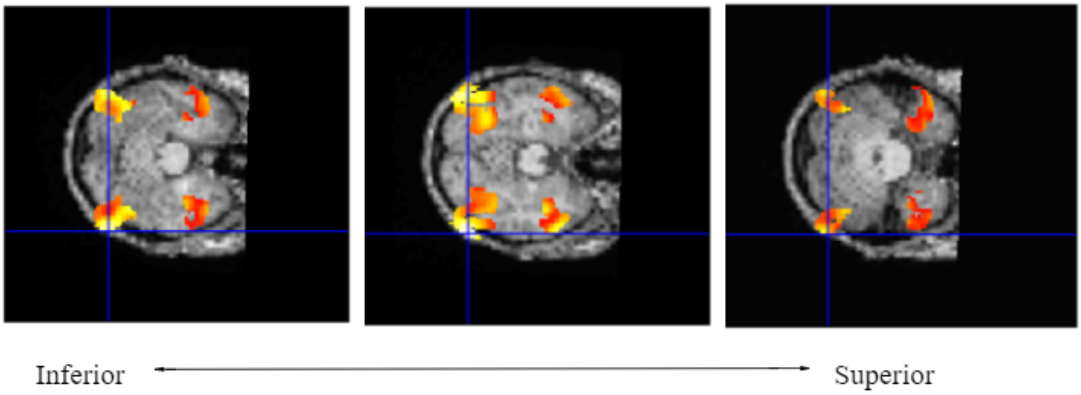


Fig. 2. FFA mask of matching face data

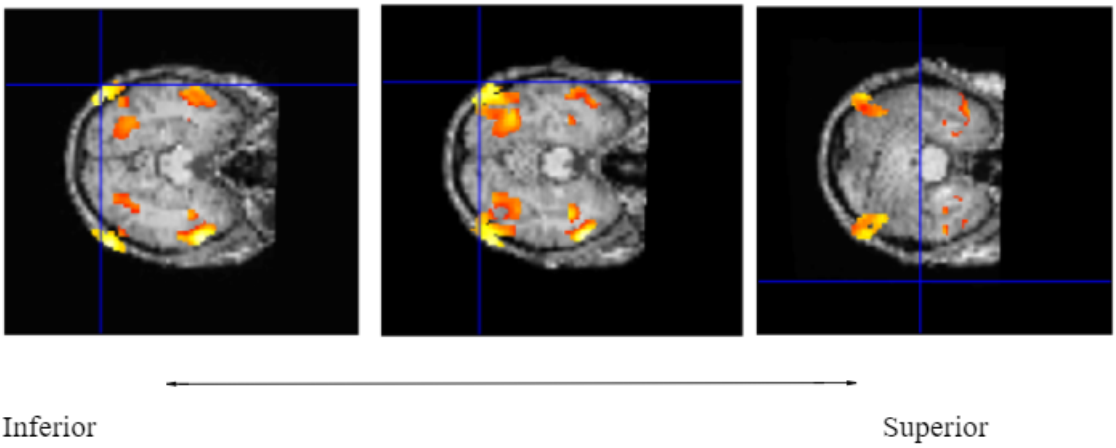


Fig. 3. FFA mask of non-matching face data

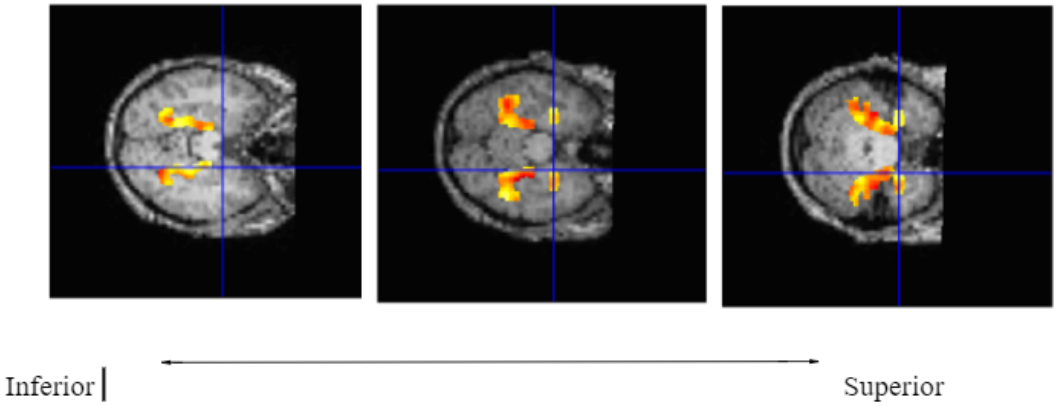


Fig. 4. PPA mask of matching nonface data

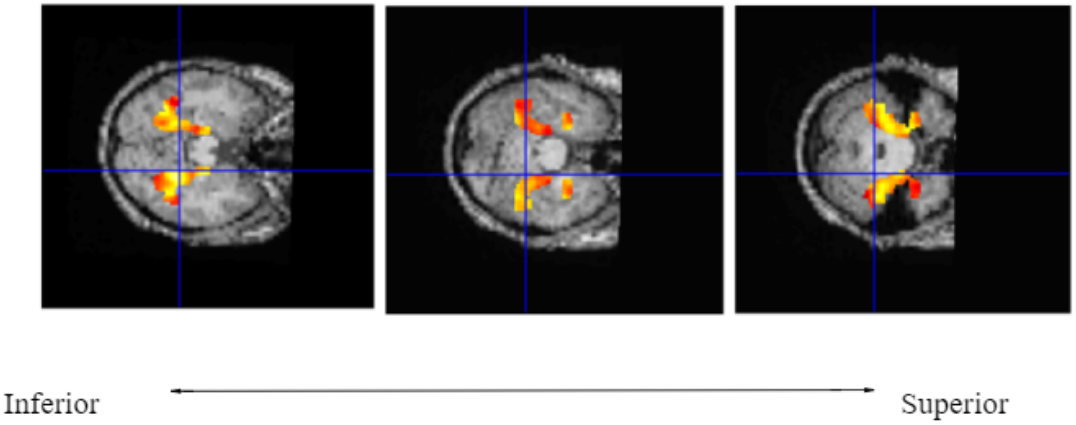


Fig. 5. PPA mask of non-matching nonface data

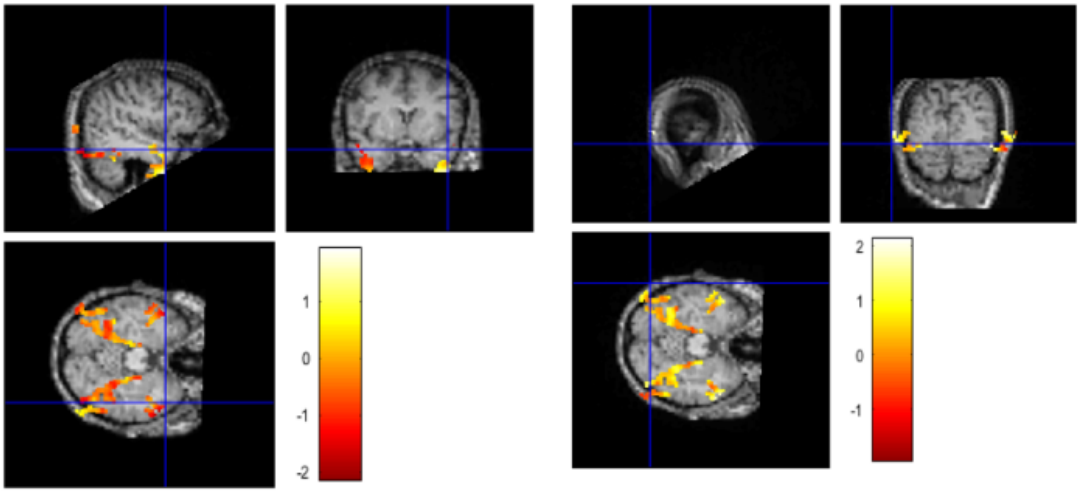


Fig. 6. An illustration of the addition of PPA and FFA masks on matching(left) and non-matching(right) face and nonface data

Section 7. Choosing the Appropriate Statistical Test for Our Hypothesis

As per the previous discussions, in this experiment we are faced with four distinct conditions and the brain response through each of them, which we will be comparing in pairs. Thus, at any instance, we will be looking at only two variable vectors (consisting of each condition's Beta weights) that also happen to be independent. Because of this, we can simply use a **paired-sample t-test,** working with the subtraction of the Beta weight vectors of matching or non-matching trials, for each category of stimuli.

In summary, in each of the following analyses, a paired sample t-test has been used for investigating the null hypothesis of no statistical difference between the values of matching and non-matching vectors. In other words, H0 is a zero mean for the result of the non-matching vector minus the matching vector..

7. 1. Analyzing the Presence of Repetition Suppression in the Case of Matching and Non-matching Face Trials

In the figures below, we can see the numerical results of the t-test executed between the *MeanBeta_MatchFace and MeanBeta_NonMatchFace* vectors. a histogram of the difference in vectors while fitted with a sample t-distribution, and finally the depiction of the t-stats of the t-test with illustrations of the mean of difference of vectors, the critical t-value, and the Confidence Interval of mean.

It can be seen from fig. 7, that h =1 and thus the Null Hypothesis is rejected in favor of the alternative. Meaning that there is in fact a significant statistical difference between the means of two vectors, or generally between the two cases of matching and non-matching trials, with the non-matching trials having a greater value (or the t-distribution being right-tailed). Also, because of the extreme magnitude of the t-value, and for better illustrations, t-value has been omitted from fig. 9.

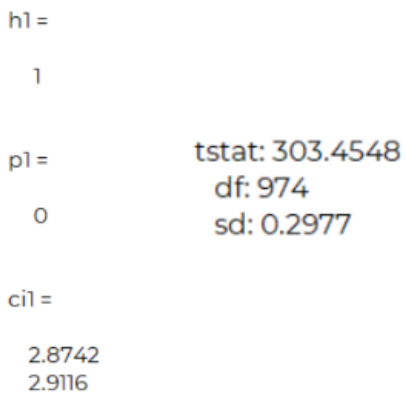


Fig. 7. Results of the t-test

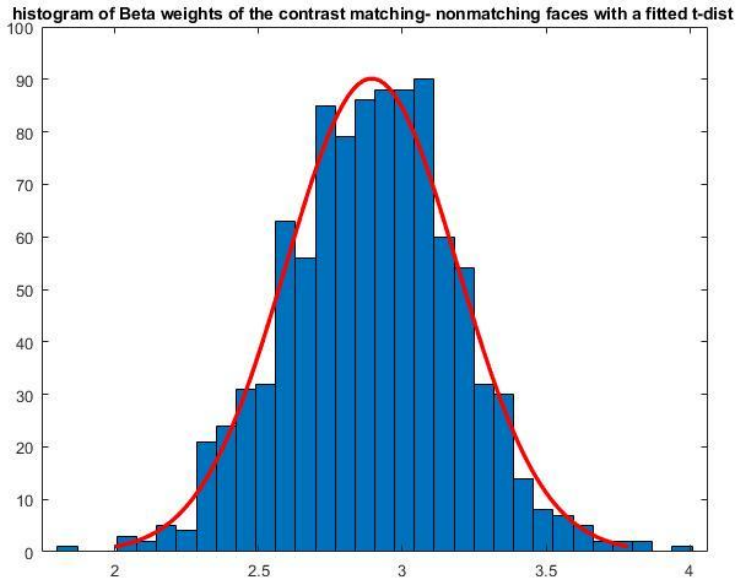


Fig. 8. Histogram of the subtraction of Beta vectors for face stimuli

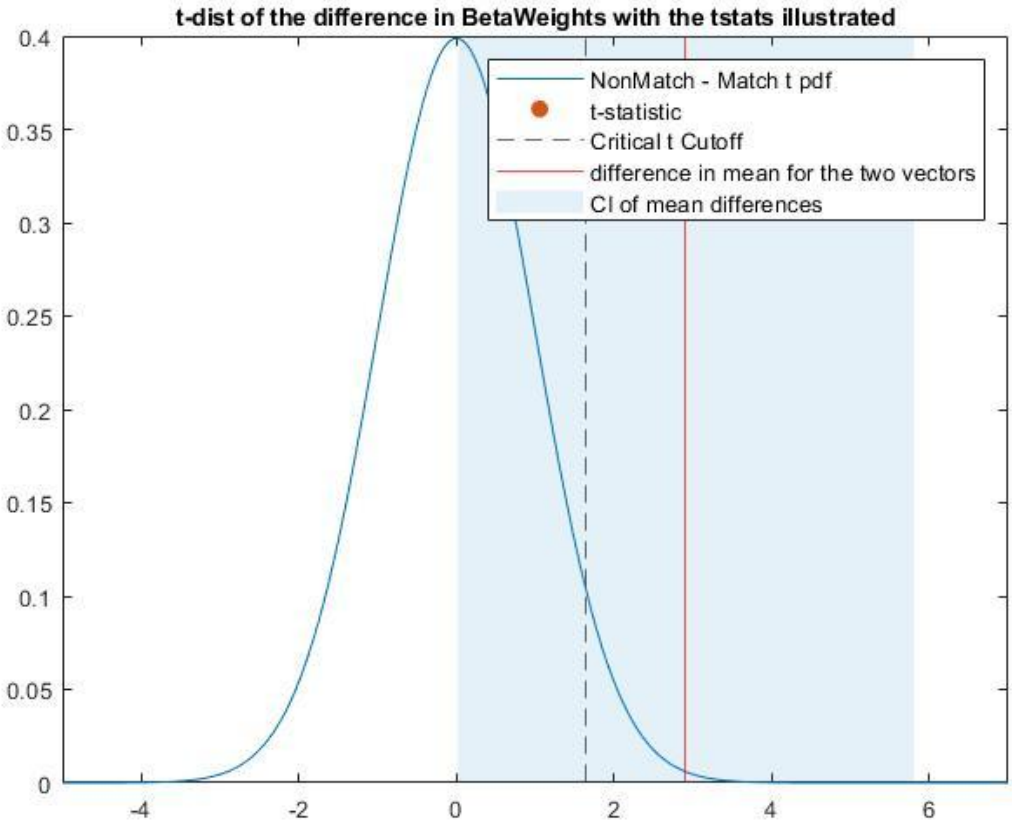


Fig. 9. Illustration of t-test stats

7. 2. Analyzing the Presence of Repetition Suppression in the Case of Matching and Non-matching NonFace Trials

Much like the previous case, again we have reported the numerical results of the t-test executed between the *MeanBeta_MatchNonFace and MeanBeta_NonMatchNonFace* vectors. a histogram of the difference in vectors while fitted with a sample t-distribution, and finally the depiction of the t-stats of the t-test with illustrations of the mean of difference of vectors, the critical t-value, and the Confidence Interval of mean.

It can be seen from fig. 10, that h =1 and thus the Null Hypothesis is rejected in favor of the alternative. Meaning that there is in fact a significant statistical difference between the means of two vectors, or generally between the two cases of matching and non-matching trials, with the non-matching trials having a greater value (or the t-distribution being right-tailed). Also, because of the extreme magnitude of the t-value, and for better illustrations, t-value has been omitted from fig. 12.

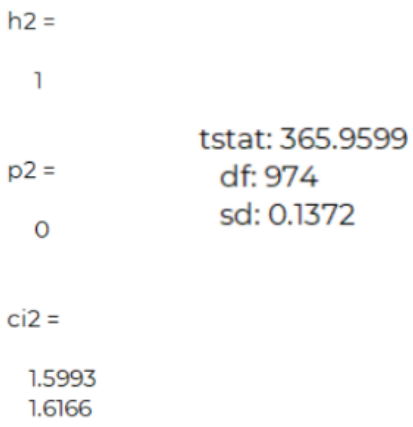


Fig. 10. Results of the t-test

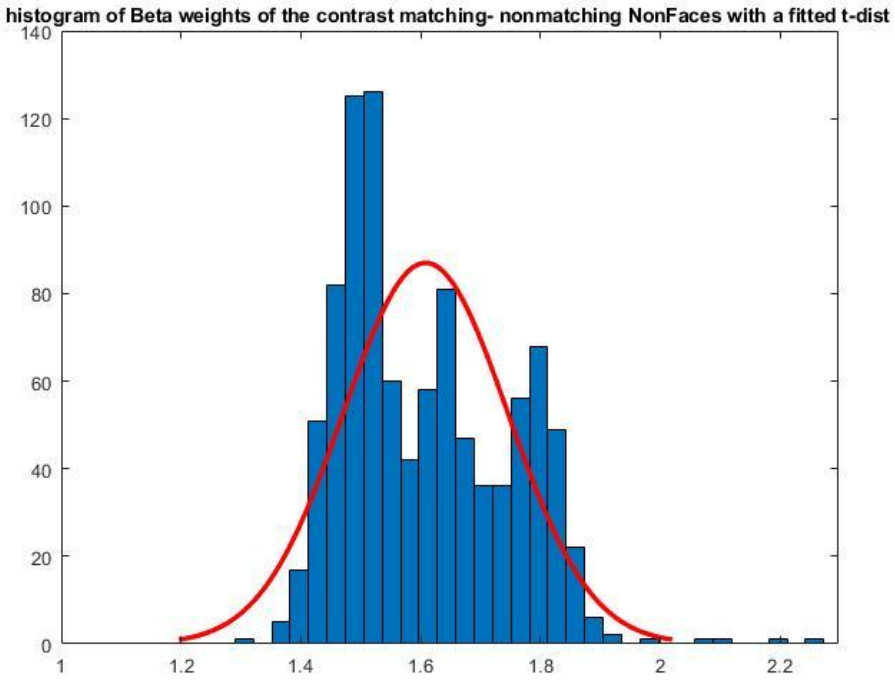


Fig. 11. Histogram of the subtraction of Beta vectors for nonface stimuli

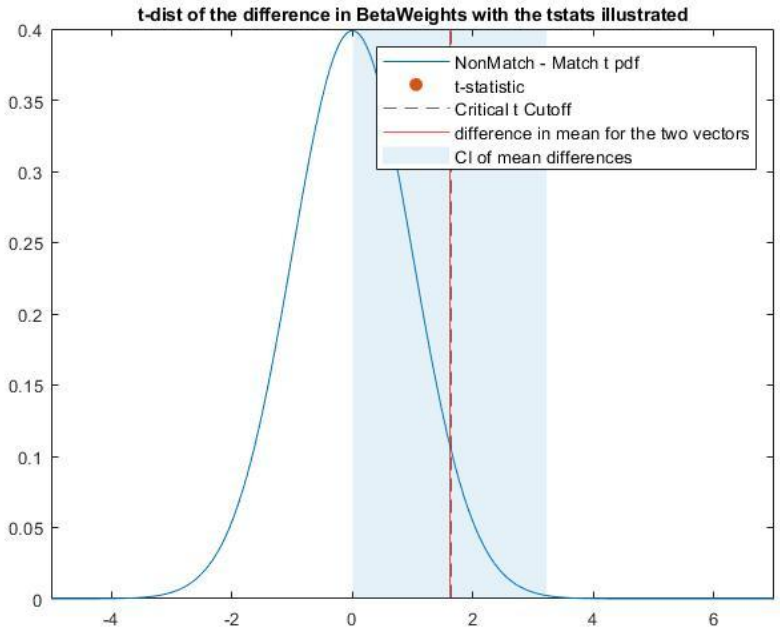


Fig. 12. Illustration of t-test stats

7. 3. A Deduction on the Scale of Repetition Suppression Compared Among PPA and FFA

As an extra step, we investigate the scale of the RS effect in FFA and PPA through the following steps.

First we make the **secondary guess** that the magnitude of RS is greater in FFA than in PPA. For investigating this question, we propose the **Null Hypothesis** that there is no significant statistical difference between the means of the Beta weight vectors, obtained from subtracting matching trials' Beta vector from the non-matching trials' Beta vector, among face and nonface data.

For testing this Null Hypothesis, we first subtract the Beta weights of the two conditions in each of the PPA and FFA stimuli sets, and then check to see whether or not this difference or this supposed RS is more significant in value in the PPA sets, or the FFA sets. The Null Hypothesis here would be that there is no difference between the mean of these two newly obtained vectors, and that PPA and FFA have an almost similar scale of RS. A **right-tailed t-test** is used with an inclination towards the FFA RS being greater than the PPA's.

The numerical results of the t-test executed between the *MeanBeta_NonFace and MeanBeta_Face* vectors, a histogram of the difference in vectors while fitted with a sample t-distribution, and finally the depiction of the t-stats of the t-test with illustrations of the mean of difference of vectors, the critical t-value, and the Confidence Interval of mean, have been sequentially reported in figures 13, through 15. It can be seen from fig. 13, that h =1 and thus the Null Hypothesis is rejected in favor of the alternative. Meaning that there is in fact a significant statistical difference between the means of two vectors and more specifically that the first vector belonging to the FFA data, has a greater mean than that of the PPA data. Also, because of the extreme magnitude of the t-value, and for better illustrations, t-value has been omitted from fig. 15.

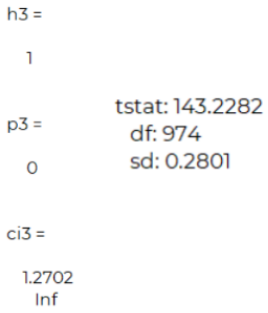


Fig. 13. Results of the t-test

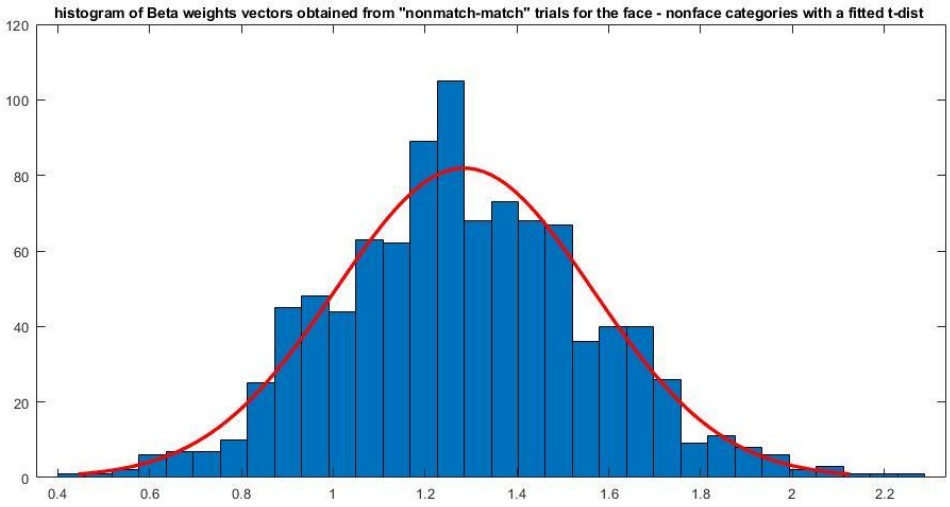


Fig. 14. Histogram of the subtraction of Beta vectors for face and nonface stimuli

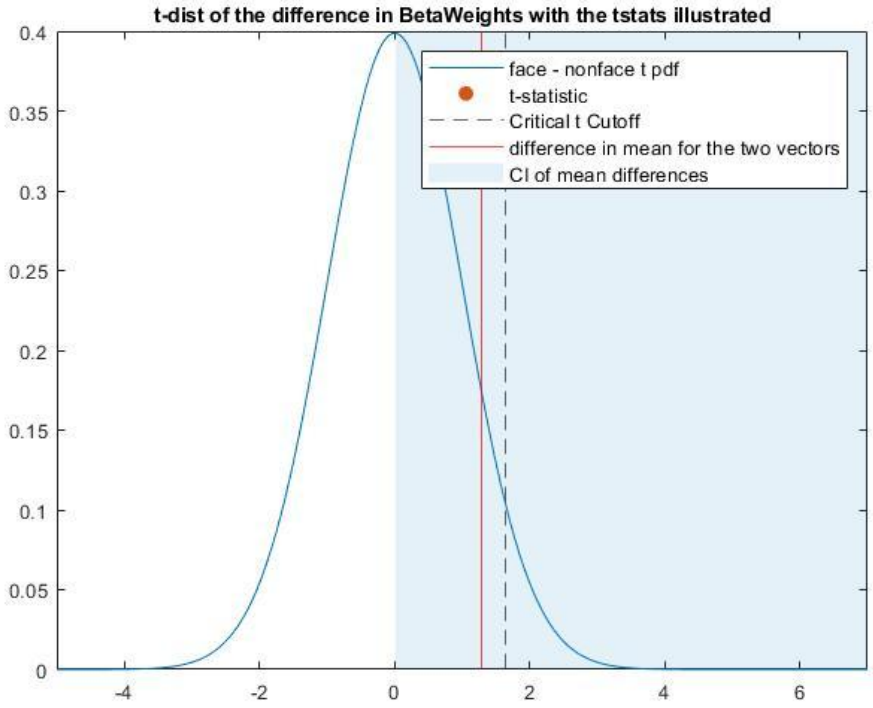


Fig. 15. Illustration of t-test stats

Section 8. Conclusion

To summarize, we proposed a research question consisting of two parts: firstly, the presence of Repetition Suppression in the brain while faced with matching or non-matching trials. Secondly, we guessed that supposing that there actually is a Repetition Suppression effect in the brain, then it must be larger in the FFA than it is in the PPA.

For testing these hypotheses, we employed simple one way and two way paired sample t-tests. These tests, and their respective illustrations, yielded the rejection of our Null Hypothesis, meaning that in fact, our initial guesses of the presence of the RS effect and its larger scale in the FFA in comparison to the PPA, stands true.